Chapter 10

Metal–ceramic crowns

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CHAPTER CONTENTS

Indications for metal–ceramic full coverage crowns 129
Metal–ceramic full coverage crowns 129
The metals 130
The metal–ceramic bond 130
The ceramic 131
Advantages of metal–ceramic full coverage crowns 131
Disadvantages of metal–ceramic full coverage crowns 131
Patient consent 132
General principles of tooth preparation 133
Occlusal reduction 133
Axial reduction, retention and resistance form 134
Cervical finishing lines 134
Finishing of preparations 134
Posterior metal–ceramic full coverage crown 134
Occlusal preparation 134
Buccal preparation 134
Lingual preparation 136
Approximal preparation 136
Anterior metal–ceramic full coverage crown 137
Incisal preparation 137
Labial preparation 137
Palatal/lingual preparation 137
Approximal preparation 139
Laboratory prescription 139
Summary 140

INDICATIONS FOR METAL–CERAMIC FULL COVERAGE CROWNS

If a tooth requires a full coverage indirect restoration and there is a need for an aesthetic restoration, a metal–ceramic crown can be considered. However, all-ceramic restorations may be more appropriate in patients with extremely high aesthetic demands and these are increasingly being used by some clinicians. There are also special circumstances that require specific modifications to the standard metal–ceramic crown detailed throughout this chapter. Metal–ceramic restorations can be waxed-up or milled to provide space for a movable component of a fixed prosthesis (Figure 10.1; see also Chapter 19) or a rest for a removable partial denture (see Chapter 8). These must not be confined to ceramic as this can lead to catastrophic fracture and failure. In addition, guide planes and undercuts to aid the retention of a removable prosthesis can be incorporated into the restoration.

METAL–CERAMIC FULL COVERAGE CROWNS

Metal–ceramic crowns consist of:

- A metal substructure
- An opaceous ceramic layer
- A veneering ceramic layer (Figure 10.2).

The rationale for this restoration is to combine the strength of a metal substructure with the aesthetic qualities of dental porcelain. However, tooth preparation for

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Figure 10.1 Female component of movable connector that is incorporated into a fixed prosthesis. The component is confined to metal within the restoration.

a full coverage metal–ceramic restoration is destructive of tooth substance (Figure 10.3) as clearance needs to be created not only for the metal but also for the opaceous ceramic and the dentine and enamel effect veneering ceramics. To provide sufficient strength the metal substructure should normally be between 0.3 and 0.7 mm thick; ideally this should be at least 0.5 mm thick with a greater thickness on the occlusal surface.

The metals

Metal alloys used in dentistry fall into three groups:

- *Gold alloys*. These have been described in the previous chapter.
- *Base metal alloys*. These are usually alloys of nickel and chromium, some with small amounts of other metals (e.g. molybdenum). Some patients may be allergic to nickel and these alloys should be avoided in such patients.
- *Titanium alloys*.

Any of these metals can be used for metal–ceramic restorations, providing their melting point is above the sintering temperature of the ceramic. If the two are too close, thin sections of the metal might melt and the metal framework may warp or sag when it is placed in the ceramic furnace.

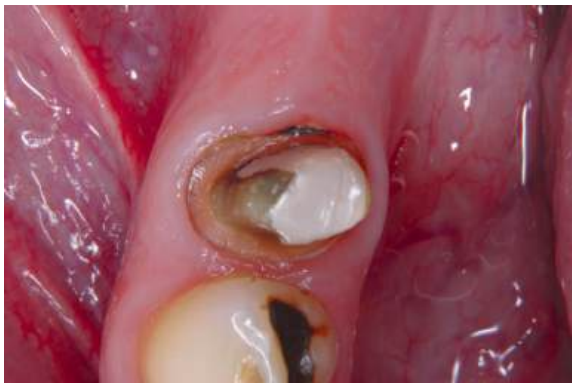


Figure 10.3 Metal–ceramic crown preparation with shoulder buccally merging into a chamfer lingually. Reduction of the buccal cusp has significantly reduced the amount of remaining tooth tissue.

The price of metals fluctuates and therefore prescribing patterns often vary depending on the cost of the constituents of these metals. Precious metal alloys have a long clinical track record with good longevity in longitudinal studies. Titanium alloys have recently been used with some short-term clinical success; however, their long-term effectiveness remains unproven.

The metal–ceramic bond

The opaceous ceramic layer masks the colour of the underlying metal and should be 0.2–0.3 mm in thickness. Once the metal substructure has been cast, several processes prepare the metal surface to enable reliable bonding of the ceramic layers:

- Smoothing is completed in one direction, usually using a brown stone.
- The coping is sandblasted (usually with 50 μm aluminium oxide particles) and cleaned using a steam cleaner.
- Formation of an oxide layer. The exact process is often specific to the alloy used due to the varying amounts of base metal alloy present. The oxide layer can usually be formed by placing the substructure back into the furnace to complete a de-gassing (oxidization) cycle.



Figure 10.2 The making of a metal–ceramic crown, showing the metal substructure (cast using the lost wax technique) and the opaceous ceramic layer (left) and the veneering ceramic layer in its unglazed state (right).

The mechanism(s) by which the ceramic bonds to the metal coping has been a matter for debate. However, it is now generally accepted that the total bond is created by different mechanisms, with the chemical bonding thought to be most important. The total bond between the ceramic and metal substructure is thought to consist of:

- *Van der Waals forces*. The forces that attract (or repulse) molecules, but where no electrons are exchanged between the molecules (covalent bonding).
- *Mechanical interlocking*. The coping surface will have small irregularities on its surface. These provide a surface in which micromechanical interlocking, or retention can occur. The surface of the metal is often treated (e.g. sandblasted with Al_2O_3) to increase the surface area available for bonding.
- *Compression bonding*. The metal coping should expand slightly more than the overlying ceramic when heated to place the ceramic interface under compression when cooled down. In addition, the ceramic will shrink more than the metal once sintered and cooled, thereby allowing the ceramic to 'wrap-round' the metal coping.
- *Chemical bonding*. Thought to be created by the partial dissolution of the porcelain at the metal–ceramic interface causing saturation of the porcelain with metal oxides and the subsequent intimate atomic contact between the metal and porcelain.

The ceramic

The types of dental porcelain available are discussed in Chapter 11 and are used to provide the aesthetic component of these restorations. Usually glass ceramics are used for metal–ceramic restorations; these ceramics must be thermally compatible with the specific alloy being used. The difference in coefficient of thermal expansion between the ceramic and metal is important; if the difference is too great, stresses will be introduced into the ceramic, leading to crazing, cracking and ceramic fracture. Ideally, the coefficient of thermal expansion of the metal should be slightly higher than that of the ceramic (see compression bonding above). The thickness of the veneering ceramic should allow the dental technician to match the colour of the restoration to the desired shade (Figure 10.4). However, it should not be more than about 1 mm thick as the ceramic has a relatively low flexural strength and may fracture if not adequately supported by the metal framework, although newer ceramics have improved physical properties.

Advantages of metal–ceramic full coverage crowns

- *Strength*. Laboratory studies have shown metal–ceramic restorations to be stronger than conventional all-ceramic restorations. However, it must be borne in mind that bespoke all-ceramic crown materials (e.g. zirconia-based restorations) have improved physical properties compared to the conventional porcelains.
- *Fit*. Different materials produce crowns with different marginal gaps. In laboratory studies, high gold alloys often produce restorations with the smallest gaps.
- *Metal lingual and occlusal surfaces*. This reduces the amount of tooth preparation required. It is important, where possible, that occlusal surfaces are covered in metal only, not only to reduce the amount of tooth reduction required but also to reduce the potential adverse effects of having a ceramic occlusal surface: ceramic that has not been adequately polished or glazed, or subsequently been ground/worn, has been shown to accelerate the wear of the dentition opposing the ceramic. Additionally, increased plaque accumulation and a reduction in the strength of the restoration have been found following wear/grinding of the ceramic. For some patients the appearance of a metal occlusal surface may be unacceptable; for example, a lower premolar crown may be very visible and a pragmatic approach is required provided the patient is fully informed of the potential pitfalls. Metal occlusal surfaces are generally better accepted in the maxilla than the mandible where they tend to be visible.



Figure 10.4 Shade match of a metal–ceramic restoration placed on the upper left lateral incisor.

Disadvantages of metal–ceramic full coverage crowns

- *Extensive tooth preparation*. All full coverage preparations and their restorations have the potential to be the final insult to a stressed pulp, resulting in loss of vitality. The rate at which this occurs is a matter for debate (see Chapter 3). Systematic reviews of the literature have demonstrated the single crowns have a lower incidence of pulp necrosis than fixed bridgework. It must be stressed that many of these teeth may have become non-vital without crown preparation due to the various carious and restorative

insults over a prolonged number of years. It has been estimated that an anterior metal–ceramic crown preparation removes 72% of the remaining tooth volume by weight (Edelhoff & Sorensen, 2002).

- *Fit*. Metal–ceramic restorations tend to have larger marginal gaps when compared to all-metal crowns. These gaps are related to the finish line configuration and internal line angles. However, the clinical significance of these gaps in relation to caries, periodontal disease and sensitivity has never been established. However, good clinical practice is to make the marginal gap as small as possible to eliminate any potential adverse effects.
- *Aesthetics*. If a metal collar has been used, the metal substructure may be obvious at the cervical margin, either immediately or over time (Figure 10.5A). The metal and opaque ceramic layer may make it difficult to attain the translucency of natural teeth (Figure 10.5B). In addition, the value (brightness) of these restorations is often higher than that of natural teeth, reducing their aesthetic integration (Figure 10.5C).
- *Cost*. In addition to clinical time, the cost both in terms of the monetary value of the metal alloys and ceramics and the technical time (waxing-up, casting and surface treatment of a metal substructure, in addition to ceramic build-up and finishing of the ceramic layers) is considerable for these restorations.
- *Substructure design*. The metal substructure has to be designed to support the veneering ceramic, otherwise the ceramic may fracture (Figure 10.5D).
- *Strength*. Due to their high strength the remaining coronal tooth structure may fracture before the restoration following direct trauma.

PATIENT CONSENT

The longevity and possible complications associated with full coverage restorations need to be discussed together with possible alternatives (partial coverage crowns, bonded direct restorations). Ideally metal–ceramic crowns should be prescribed with metal occlusal and lingual/palatal



Figure 10.5 Problems of metal–ceramic restorations. **A** Metal margin exposure and discolouration. **B** Poor translucency, shape and shade of metal–ceramic crown on the upper right central incisor. **C** High value (brightness) of metal–ceramic restorations on the central incisors. **D** Ceramic fracture due to increased occlusal forces (bruxism) and incorrect metal coping design.



Figure 10.6 Metal–ceramic full coverage crown. The occlusal view demonstrating metal occlusal surface with small ceramic overlay for aesthetic reasons (left) and buccal view showing buccal ceramic facing prior to excess cement removal (right).

surfaces (Figure 10.6) for reasons outlined earlier in this chapter. Patients should be informed of this and the fact that they contain a metal alloy as part of the consent process. If a patient wishes a full coverage crown to be wholly tooth coloured or metal-free, then the advantages and disadvantages of all-ceramic crowns should also be addressed (see Chapter 11).

GENERAL PRINCIPLES OF TOOTH PREPARATION

Many of the general principles underlying the preparation of posterior teeth for metal–ceramic crown preparations are similar to those for full coverage gold (metal) crowns. The salient details that are common will be reiterated in this chapter to avoid excessive cross-referencing and to enable this chapter to be read independently; emphasis will be placed where they differ. Preparation of anterior teeth for metal–ceramic crowns is slightly different and this will be described separately in this chapter.

The burs used for a metal–ceramic crown preparation are the same as those used for any other crown and bridge preparation (see standard bur kit, Figure 8.3, Chapter 8). Ideally a preoperative matrix of the tooth to be prepared should be made. This matrix can be made either from a diagnostic wax-up or from the tooth once the core has been built up to the correct anatomical contours (Figure 10.7). It should be noted that the amount of tooth reduction is not necessarily the same as the clearance that may be required. For example, if the incisal length of a maxillary central incisor is shorter than the contralateral tooth, less tooth reduction will be required than if it was the same length as the contralateral tooth to achieve the desired 1.5–2 mm of clearance incisally.



Figure 10.7 Putty matrix adapted to the teeth by hand and sectioned with a scalpel blade (out of the mouth) midway over the tooth to be prepared. This can be reinserted intermittently throughout the tooth preparation to assess the amount of tooth reduction (see Figure 10.17B).

Occlusal reduction

Depth grooves are used by many clinicians to guide the amount of reduction where the original contours of the tooth and core are correct. As with all depth cuts, care must be taken when removing the 'islands' of tooth tissue between the depth cuts to ensure that the bur does not 'bounce' from one groove to another, resulting in excessive tooth substance removal at the base of each groove and possibly inadequate reduction at their peaks, leaving a rippled finish. It can be advantageous to remove the islands of tooth tissue between the grooves with the bur held at an oblique angle to the grooves (see Figure 8.8, Chapter 8).

Axial reduction, retention and resistance form

Reduction of the axial walls should ensure appropriate clearance for the restoration. There is often a temptation to tip the bur away from the long axis of the tooth to gain the ideal finish line and depth cervically; however, this often leads to an undercut being formed and insufficient tooth reduction in more coronal areas. The overall preparation configuration must be considered and will vary between clinicians and in different areas in the mouth. It is important to consider retention form and resistance form as for the full gold crown preparation (see Chapter 8).

Cervical finishing lines

The gingival contour of a natural tooth is not flat but scalloped in its horizontal orientation and therefore a crown preparation should follow this contour. The amount of scallop depends upon the gingival biotype (thin, type I biotypes have a more pronounced scallop) and other factors such as previous subgingival restorations. The finish line almost without exception should finish on natural tooth throughout its circumference, ideally with a 2 mm collar of dentine being present to provide the ferrule effect. A chamfer finishing line should be present on the palatal/lingual aspects of a preparation whilst the labial/buccal aspects should be a rounded shoulder.

Finishing of preparations

When completed, the preparation should not have sharp angles or ridges present at internal line angles or at the preparation margin. A fine-grit, tapered diamond bur or a tungsten–carbide finishing bur may be used to smooth these irregularities.

POSTERIOR METAL–CERAMIC FULL COVERAGE CROWN

The general principles of tooth preparation for a metal–ceramic crown have been outlined in the preceding section; however, this section gives a step-by-step guide to the preparation of a posterior tooth. The subsequent section will deal with anterior teeth.

The preparation should combine the relevant aspects of a full gold crown preparation for the lingual and occlusal surfaces with those of a metal–ceramic crown preparation for the buccal surfaces and buccal cusps. The sequence of tooth preparation is decided by each individual; however, the authors recommend the following.

Occlusal preparation

The occlusal reduction is carried out in a similar manner to that of a full coverage gold crown. It is commenced by making depth-orientation grooves using a parallel-sided bur to guide the depth of the grooves, ensuring that the cuspal inclines are followed. The amount of reduction depends on several factors:

- Where ceramic coverage is required for aesthetic reasons there should be 1.5–2 mm of reduction to provide clearance for an adequate bulk of both metal and ceramic. In most situations the buccal cusps of both maxillary and mandibular teeth require this reduction. Where metal-only occlusal coverage is planned, a reduction of 0.5–1 mm is required.
- The concept of the functional cusp and increased occlusal forces is viewed by some to need greater tooth preparation to provide a larger bulk of material. If metal is covering the functional cusp, then 1.5 mm of reduction is required whereas 2 mm is needed if the functional cusp is covered by both metal and ceramic. In a Class I occlusion, the functional cusps are the palatal cusps of maxillary teeth and buccal cusps of mandibular teeth. This is reversed if there is a cross-bite. The rationale for the functional cusp bevel has never been proven but is supported by some clinicians. It is unlikely ever to be proven through clinical research and so remains a contentious issue. When the tooth reduction needed for a functional cusp bevel may weaken or reduce retention, careful thought should be given to its incorporation into a preparation.

The preparation of the occlusal surface is continued by removing tooth tissue between the depth grooves, again ensuring that the outline shape of the occlusal surface is followed. The occlusal surface should then be smoothed to remove any sharp angles and depressions. It is helpful to prepare one half of the occlusal surface first and compare the prepared half to the other to aid the clinician in the assessment of the correct amount of clearance. Alternatively, a sectioned matrix can be used to verify the amount of clearance. The stages of occlusal reduction are shown in Figure 10.8.

Buccal preparation

Buccal reduction should be completed after placement of depth grooves in at least two planes to mirror the buccal curvature of the tooth (Figure 10.9). Grooves should be placed to a depth of at least 1.5 mm to create clearance for both the metal and ceramic. The finish line should ideally be supragingival and be a rounded shoulder or a deep chamfer. For aesthetic reasons, however, the finish line is often placed just into the gingival sulcus.

A round-ended bur is often used to reduce the buccal surfaces and create the approximate finish line configuration



Figure 10.8 Occlusal reduction for a metal–ceramic crown. **A** Occlusal depth grooves. **B** Occlusal reduction. **C** Functional cusp reduction and bevel.

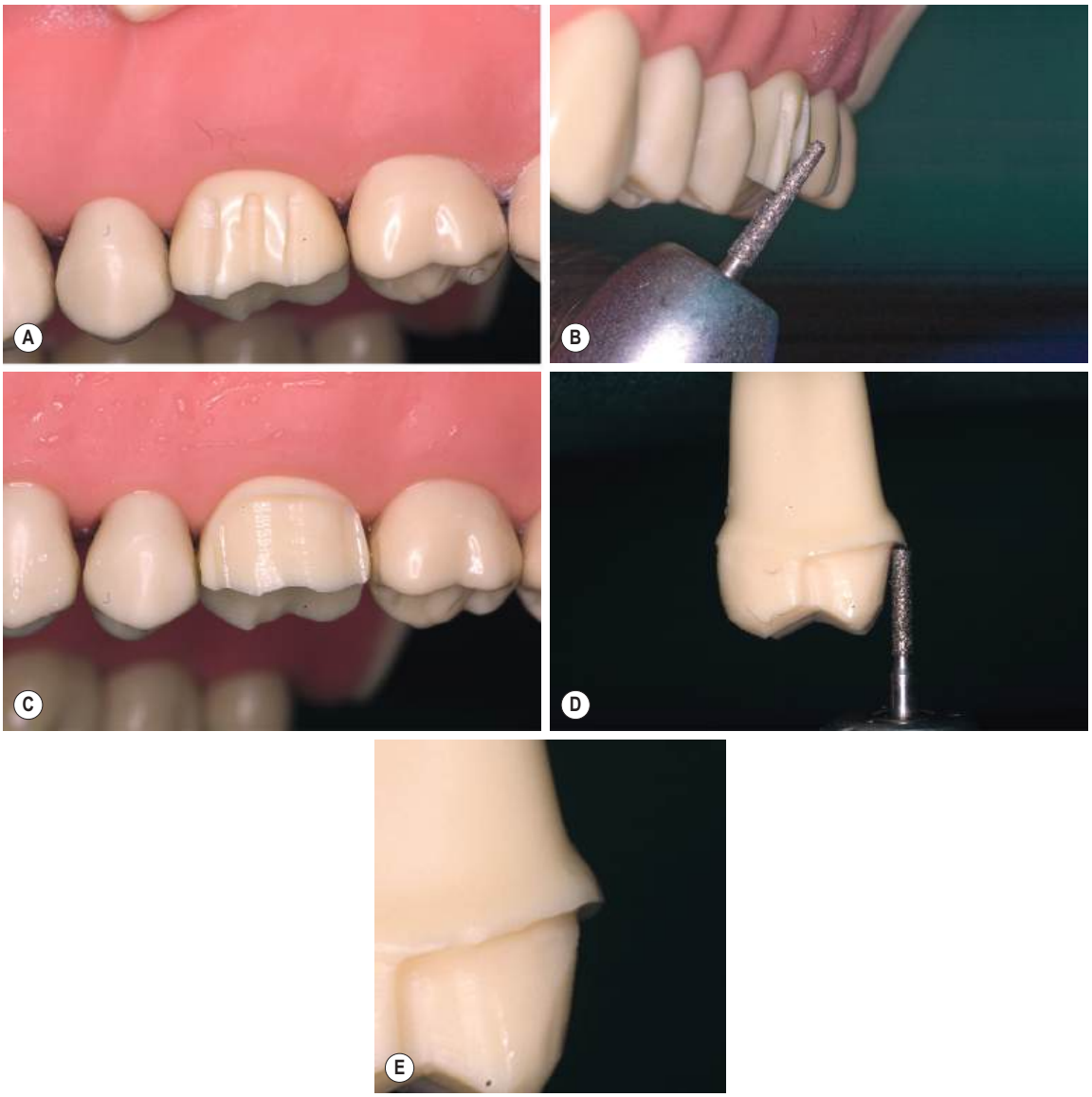


Figure 10.9 Buccal preparation for a metal–ceramic crown. **A** Buccal depth grooves parallel to the long axis of the tooth. Remember that at least two planes of buccal reduction are required. **B** Second plane of buccal reduction. **C** Buccal reduction with a supragingival finish line. **D** A round-ended bur may create a lip of enamel; observe the gap under the flat-ended bur. **E** A flat-ended bur allows for easier preparation of the desired finish line.

(Figure 10.9C). It should be noted that it is difficult to create a rounded shoulder finish line using a chamfer/torpedo bur alone as their shape will often leave a lip of enamel/dentine (Figure 10.9D). This lip is prone to fracture either intraorally or on the laboratory model, both of which will provide a restoration with an unacceptable marginal fit. A flat-ended bur should be used to finalize the desired margin (Figure 10.9E).

Lingual preparation

The lingual reduction is completed in the same manner as described for the full gold crown (Chapter 8). This provides a chamfer margin of approximately 0.5 mm depth. A chamfer or round-ended tapered bur is used to half its width, thus creating a margin with the correct shape and amount of reduction (Figure 10.10).

Approximal preparation

The buccal and lingual margins need to be joined in the approximal areas. Contact points are completely removed, creating separation of the teeth. It is important that adjacent teeth are not damaged at this stage and therefore a narrow, tapered bur is used in a bucco-lingual direction. The bulbosity of some teeth can be used to reduce the chance of damaging the adjacent teeth whilst maintaining the taper and finish line of the preparation (Figure 10.11). When separation is complete a chamfer/round-ended tapered bur is used to complete the approximal preparation in a similar fashion to that described for the lingual chamfer.

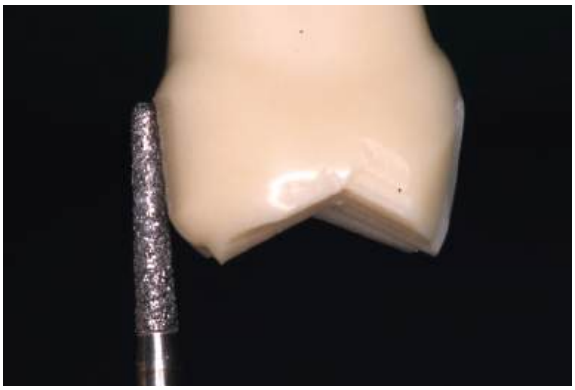


Figure 10.10 Lingual reduction using half the width of a round-ended tapered bur.

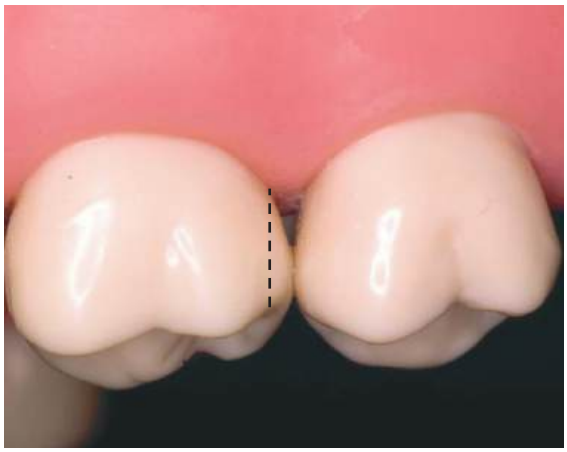


Figure 10.11 Approximal reduction. The risk of damage to adjacent teeth is aided by tooth bulbosity, the dotted line illustrates the intended wall after preparation.

There is debate as to the configuration of the approximal area. Some authors advocate a 'blending' of the buccal rounded shoulder/heavy chamfer to the smaller lingual chamfer, whilst others state a 'wing' should be present to demarcate the two marginal configurations (Figure 10.12). A 'wing' provides several advantages: firstly, it affords additional retention and resistance form that are often lacking in molar crown preparations, and secondly, the 'wing' indicates to the laboratory technician where the porcelain margin should finish. Authors that advocate the proximal 'wing' also state that this preserves tooth structure, but Figure 10.12 shows that this is minimal.

The final decision lies with the treating clinician; however, one must ensure that blending does not start too far

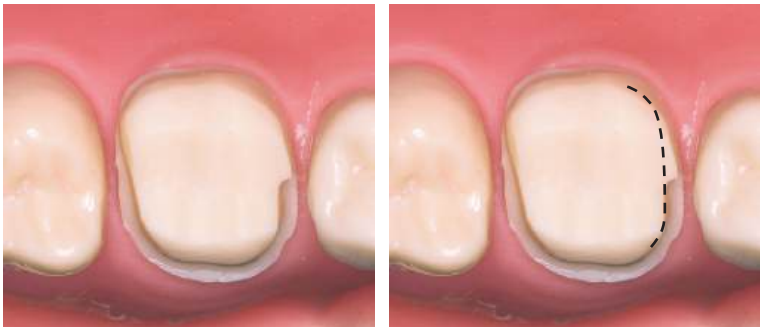


Figure 10.12 Approximal configuration. Distal wall: the buccal shoulder is 'blended' into the palatal chamfer, whereas the mesial wall has a 'wing' (left). An outline of the 'blended' margin is superimposed on the 'wing' (right), giving an indication of the relative amounts of tooth removal. It must be emphasized that this blended margin should provide the technician with sufficient space to provide an aesthetic restoration.

buccally as this will reduce the amount of clearance for metal and ceramic and may result in compromised aesthetics. Figure 10.13 demonstrates a crown with poor aesthetics on the mesio-buccal aspect of the second premolar tooth due to inadequate reduction; it is unknown what marginal configuration was present in this area.

Prior to making the provisional restoration and impression, the geometry of the preparation should be reassessed (Figure 10.14). Figure 10.15 shows typical metal–ceramic preparations on posterior teeth, with blended approximal margins.



Figure 10.13 Inadequate reduction of the mesio-buccal aspect (arrowed) of the upper left second premolar tooth resulting in poor aesthetics due to the opaceous porcelain not being adequately masked by the veneering ceramic.

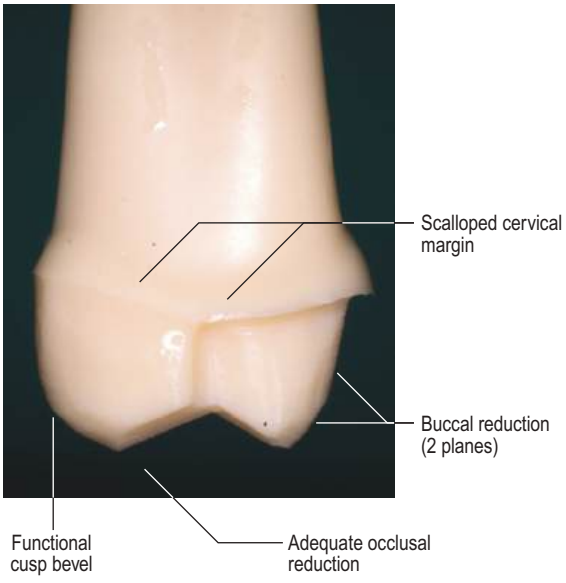


Figure 10.14 Overview of completed metal–ceramic preparation for posterior teeth.



Figure 10.15 Typical metal–ceramic preparations of the posterior teeth in the upper right quadrant. Inset: the metal–ceramic crowns in situ.

ANTERIOR METAL–CERAMIC FULL COVERAGE CROWN

Incisal preparation

The incisal reduction is commenced by making depth-orientation grooves (Figure 10.16A) using a parallel-sided bur to guide the depth of the grooves. Reduction of the incisal edge should result in the tooth height being reduced by approximately 2 mm. The grooves should then be connected to each other (Figure 10.16B) to produce a uniform reduction of the incisal edge. The preoperative matrix can also verify the incisal clearance.

Labial preparation

The labial reduction should be in two or three different planes to allow the technician to create a restoration with the ideal emergence profile, contour and shape, and to match the selected shade. Depth orientation grooves and a matrix can be used to guide the clinician in the amount and location of reduction. These should result in a uniform reduction of the labial surface in the magnitude of around 1.5 mm (Figure 10.17).

Palatal/lingual preparation

The palatal reduction of anterior teeth is in two distinct stages:

- Palatal axial or palatal collar reduction (Figure 10.18A)
- Palatal cingulum reduction (Figure 10.18B).

These provide a uniform palatal reduction with 0.5 - 1 mm of occlusal clearance and a chamfer margin. The palatal

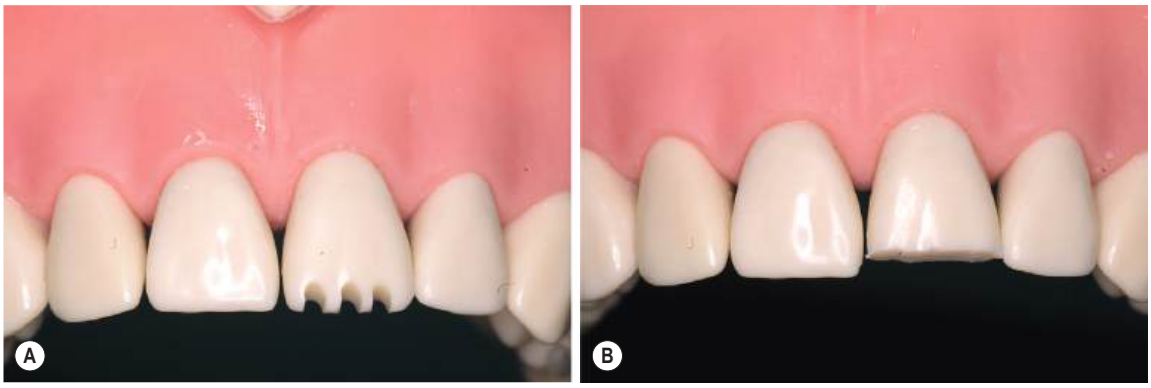


Figure 10.16 Incisal preparation for a metal–ceramic crown. **A** Depth grooves on incisal edge. **B** Uniform reduction of incisal edge.

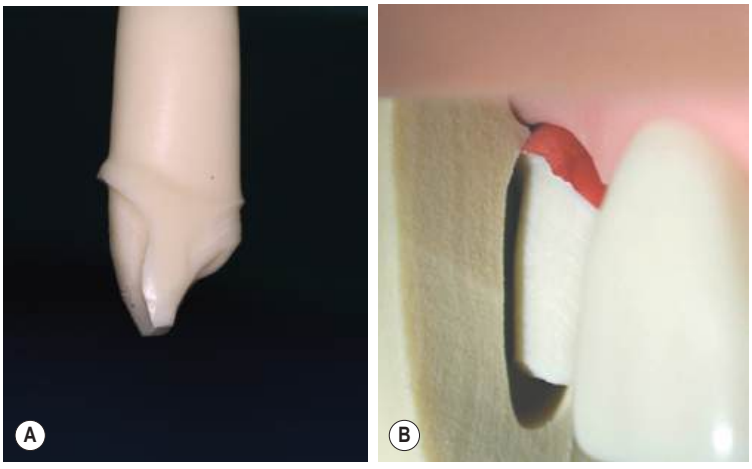


Figure 10.17 Labial preparation for a metal–ceramic crown. **A** Note uniform reduction of labial face in at least two planes. **B** Reinsertion of the putty matrix seen in Figure 10.7 shows adequate and uniform reduction.

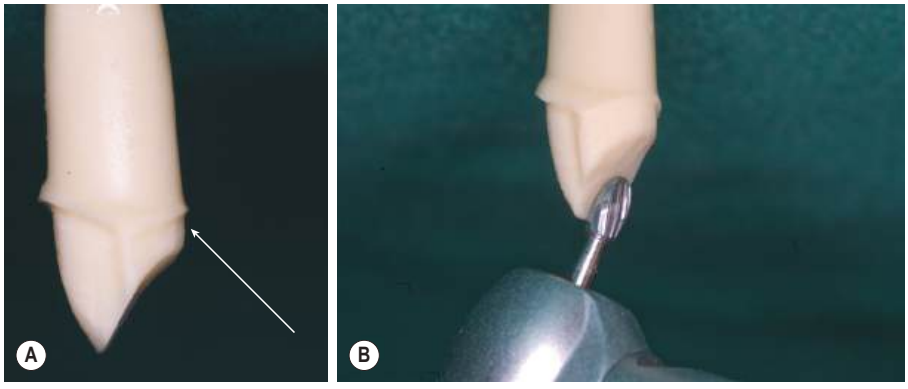


Figure 10.18 Palatal preparation for a metal–ceramic crown. **A** Stage 1: the cervical, palatal collar reduction using a round-ended tapered bur. **B** Stage 2: cingulum reduction using a round or rugby ball-shaped bur.



Figure 10.19 Approximal preparation with a 'wing' demarcating the transition from labial shoulder to palatal chamfer.



Figure 10.20 Two typical metal–ceramic crown preparations on anterior teeth. Note the labial shoulder, palatal chamfer and adequate occlusal clearance.

axial reduction should be as parallel to the cervico-labial reduction as possible to give the preparation good retention and resistance form.

Approximal preparation

The buccal and lingual margins again need to be joined in the approximal areas. The configuration of this junction has been discussed previously for the posterior preparation and these principles apply to anterior preparations (Figure 10.19). Figure 10.20 shows typical anterior metal–ceramic crown preparations.

LABORATORY PRESCRIPTION

Attention has been drawn to the fact that the quality of laboratory prescriptions is generally poor (Lynch & Allan, 2005). The basic prescription should include patient details, that a metal–ceramic crown is required, and the shade and characterization. When starting in a new practice and not knowing the technician, the importance of an accurate prescription is essential; however, as experience between the clinician and the technician improves, and each understands the other's needs, the detail of the prescription may change. Also, many technicians offer a shade-taking service which also allows them to see the tooth and so understand the situation better. In addition to the basic prescription details outlined, other pieces of information are needed to allow the technician to fabricate the desired metal–ceramic restoration.

Materials

- *Metal composition*. The classification of metal to be used has to be prescribed (precious, semi-precious, non-precious or titanium).
- *Ceramic type*. The ceramic types available are described elsewhere (feldspathic, pressable, etc.). The majority of metal–ceramic restorations utilize feldspathic porcelains, with some brands being considered as more aesthetic.

Design

- *Contact points*. The size and location of the contact point has to be considered. Contact points, from an aesthetic point of view, are found in pairs and ideally should show symmetry (e.g. the mesial surface of both upper lateral incisors). The exceptions are the contact between the upper and lower central incisor pairs. If one contact point of the pair is long and wide, the contralateral prosthetic contact should match. In the absence of a dental papilla, a long contact point may be used to improve the aesthetics by eliminating a black triangle; however, this can often be at the detriment of the emergence profile.
- *Gingival margin*. The type of gingival margin of metal–ceramic restorations has to be prescribed. It is usually either a porcelain butt finish or a metal collar.
- *Metal coverage*. The surfaces to be covered by ceramic should be described. Usual practice is to fabricate the occlusal and palatal/lingual surfaces in metal.
- *Occlusal factors*. Single crowns should provide even and holding occlusal contacts in the intercuspal position. Unless the crown is involved in guidance in lateral excursion (e.g. a premolar in group function), the posterior single crown should usually be free of contacts on both the working and non-working side in lateral excursions. If a custom incisal guidance table is not being used, then an anterior single crown should follow the anatomical contours of adjacent teeth to prevent it from providing an occlusal interference.

SUMMARY

Metal–ceramic crowns are an essential part of the dentist's armamentarium, allowing provision of high-strength aesthetic restorations with a history of longevity. The preparation of a tooth for such a restoration is, however, exacting, to allow the technician scope to provide a restoration with a good appearance and fit. This requires a very skilled technician and good three-way communication between the patient, clinician and technician.

FURTHER READING

Further details relating to the fabrication and technical aspects of metal–ceramic restorations can be found in W.P. Naylor's text, *Introduction to Metal–ceramic Technology* (Quintessence Publishing Company (Chicago), 1992) and J.W. McLean's book, *The Science and Art of Dental Ceramics* (Quintessence Publishing Company (Chicago), 1979).

Edelhoff, D., Sorensen, J.A., 2002. Tooth structure removal associated with various preparation designs for anterior teeth. J. Prosthet. Dent. 87, 503–509.

Lynch, C.D., Allen, P.F., 2005. Quality of communication between dental practitioners and dental technicians for fixed prosthodontics in Ireland. J. Oral Rehabil. 32, 901–905.